

Agostino Scalzullo

Audio Software & DSP Engineer

C++ · DSP · Max/MSP · JUCE · VST3/AU · Real-Time Audio · Hardware Integration · MIDI/MPE

London, UK · OSTIN SOLO · VSTOPIA
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PROFILE

Audio software and DSP engineer building real-time music systems across native C++, Max/MSP, JUCE plug-ins and interactive hardware. Developer of commercial OSTIN SOLO music software and founder of VSTOPIA, with hands-on experience taking systems from musical concept and rapid prototyping through native implementation, hardware integration, testing, packaging and release. Work spans DSP, MIDI/MPE, networked audio and selected machine-learning audio systems.

CORE SKILLS

Audio / DSP: C/C++, Objective-C++, JUCE, Max SDK / Max/MSP, VST3/AU, real-time DSP, MIDI/MPE, synthesis and modulation; architectural familiarity with RNBO and its C++ / WebAssembly export pipeline

Hardware / embedded: Arduino, ESP32, MCP23017 I/O expansion, firmware integration and modification, physical controls and sensors, microcontroller-to-host integration

Real-time / platform: WebRTC, Opus, RTP MIDI, WebSockets, AVFoundation, platform APIs, Swift/SwiftUI, macOS/iOS/Windows

Engineering / deployment: profiling, benchmark design, parity and regression testing, cross-platform builds, conditional compilation, model/runtime packaging, Developer ID signing and notarization

Audio ML / vision: source separation, BS-RoFormer, Demucs, MossFormer2, ECAPA, MLX, Core ML, CLAP embeddings, Depth Anything, MediaPipe, ARKit

SELECTED WORK

D_{Sm} — Dynamic Split Module

Neural separation, restoration and web-sampling environment for Ableton Live, with 63 model workflows spanning Demucs, RoFormer and related architectures. Covered by Sound On Sound and listed in the Cycling '74 project ecosystem.

Native RoFormer External & VST3/AU — Work in Progress

Native C++/MLX implementation bringing six-stem BS-RoFormer separation into a Max/MSP external and an in-development plug-in architecture. Across tested Apple Silicon workloads, optimisation achieved a **median 1.85× speedup**, reaching **2.5× on MelBand RoFormer**; native six-stem BS-RoFormer achieved **RTF ≈ 0.36 (~2.8× real time)** on M4 Pro with bit-exact parity against the Python MLX reference.

AUDIOENCE

JUCE/C++ networked collaboration plug-in architecture combining real-time audio, MIDI, WebRTC/P2P communication, Opus/PCM modes, resampling and DAW synchronisation across macOS and Windows.

SELECTED WORK — CONTINUED

Native Max / MPE / Hardware Systems

Native C++ Max externals and hardware-integration work spanning multitouch, MPE, computer sensors, Arduino/ESP32 microcontrollers, MCP23017 I/O expansion, Depth Anything and AprilTag workflows. Includes Objective-C++ bridge layers, platform APIs, firmware integration/modification, native model/runtime integration and ongoing JUCE VST3/AU migration. Earlier physical-computing prototypes included an MCP23017 encoder/button controller, 12-channel capacitive-touch MIDI control, ribbon-sensor pitch control, and phone/accelerometer-controlled servo positioning, first over an external Bluetooth module and later ESP32.

SoundCloner — Private R&D

Native synthesis and audio-analysis research platform combining deterministic DSP, oscillator/envelope/modulation systems, parameter/genome representations, similarity scoring, candidate retrieval, ML inference and large synthetic evaluation infrastructure.

Real-Time Target-Speaker Voice Isolation — Private R&D

Real-time target-speaker extraction using MossFormer2 + ECAPA, with speaker enrollment and suppression of competing voices.

Sample-Space Exploration — Private R&D

CLAP/audio embeddings, classifier training, similarity retrieval and Three.js spatial exploration of audio corpora, supported by benchmark and evaluation tooling.

iOS Computer Vision — Private R&D

Swift/SwiftUI iOS application work using ARKit, SceneKit, AVFoundation and MediaPipe Tasks Vision for real-time facial/eye landmark processing and guided physical-makeup workflows.

ENGINEERING VALIDATION

Repeatable benchmarking and regression methodology across projects: controlled baselines, latency/throughput profiling, held-out ML evaluation, parity testing, hardware-in-the-loop checks and cross-machine validation across Apple Silicon, Intel Mac and Windows execution paths where supported.

EXPERIENCE

Founder & Audio Software Developer — VSTOPIA | 2023–Present

Built an independent software ecosystem spanning product development, web/backend infrastructure, distribution, licensing, updates and developer workflows.

Independent Audio Software Developer — OSTIN SOLO | 2018–Present

Commercial and experimental audio technology across DSP, Max for Live, native Max externals, plug-ins, machine learning, computer vision, MPE/hardware control and networked music systems.

Founder, Artistic Director & Experience Designer — Macrowave / Cacao Discoteca / Basement | 2014–2016

Directed electronic-music and multimedia programmes in Italy for audiences up to 3,000, spanning artist curation, production, projection mapping, scenography and interactive installations.

EDUCATION

Music Production — SAE Institute | 2019–2020 · Recording, editing, mixing, mastering, sound design and synthesis

Electronic Music & Musicianship — Italian Conservatory | c. 2004–2009

RECOGNITION & PUBLIC EVIDENCE

Sound On Sound coverage of DS_m · Cycling '74 official project listing · OSTIN SOLO commercial product catalogue · public AUDIOENCE platform and technical documentation.

Full technical portfolio: ostinsolo.github.io/ostin-solo-portfolio/